

Introduction to Quarto

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Introduction

- Quarto provides a unified authoring framework for data science, combining your code, its results, and your prose.
- Quarto documents are fully reproducible and support dozens of output formats, like PDFs, Word files, presentations, and more.

Usage of Quarto files

- Quarto files are designed to be used in three ways:
 - i. For communicating to decision-makers, who want to focus on the conclusions, not the code behind the analysis.
 - ii. For collaborating with other data scientists (including future you!), who are interested in both your conclusions, and how you reached them (i.e. the code).
 - iii. As an environment in which to do data science, as a modern-day lab notebook where you can capture not only what you did, but also what you were thinking.

Types of Contents in Quarto

1. An (optional) **YAML Header** surrounded `---`
2. **Chunks** of R code surrounded by `````
3. Text mixed with simple text formatting like heading and italics

YAML Header

- Quarto default YAML header includes the following metadata surrounded by three dashes ---:
 - i. title
 - ii. author
 - iii. format
 - iv. editor

Formatted text

- You can format text in quarto using the following options:
 - i. `headers`: one uses `#`, `##`, `###`, `####` for one, two, three and four level headings respectively
 - ii. `bold`: place the text in between the two asterik `**` `**`
 - iii. `italics`: place the text in between one asterik as `*` `*`

Code Blocks

- Code blocks appear highlighted in gray throughout the qmd document.
- They are surrounded by three tick marks on either side (`````) in source mode with the starting three tick marks followed by curly brackets `{}` with some other code inside.
- The tick marks indicate the start of a code section, and the bits found between the curly brackets `{}` indicate how R should read and display the code.
- These are the sections where you add R code, such as summary statistics, analysis, tables, and plots.
- Supported Languages are R, Python, Bash, SQL, Rcpp, Stan, JavaScript, CSS

Execution Options

- There are a variety of options for how to handle the code and its output, called 'execution options'.
- These options are specified at the start of the code chunk, and begin with #|

Execution Options

- `eval: true` indicates that the code chunk should be 'evaluated' or 'run'
- `echo: true` indicates the code chunk should be 'echoed' or shown along in the output
- `warning: false` indicates that any warning messages should not be shown in the output

Execution Options

- `include: false` is a catch all, indicating neither the code chunk nor results should be shown in the output
- `code-fold: true` indicates that the code chunk is shown, but presented in a collapseble format
- `code-summary: "Title of folding code"` indicates the name used as the heading for the folding code

Callout Blocks

- Callouts are used to draw extra attention to certain concepts, or to more clearly indicate that certain content is supplemental or applicable to only some scenarios.
- Callouts start with `::: {.callout-note}`, then there's the content, and it ends with `:::`.
- There are five types of callouts: `note`, `warning`, `important`, `tip`, and `caution`.

Figures in Quarto

- You can produce figures in Quarto documents as shown

```
1 library(ggplot2)
2
3 ggplot(diamonds) + geom_bar(aes(x = color), width = 0.2) +
4   theme_minimal()
```